

STRATEGIC COST ANALYSIS

REVIEW SESSION – EMBA 60

Question 1 - Panda Flowers

Panda Flowers is developing 50 Hectares of Green Houses for Roses using Spanish Technology. 40 Hectares are already producing Roses which are exported to USA.

Panda Flowers is situated in the Flower Business about 100 Km from Nairobi, the capital of Kenya.

-From Panda Flowers' website (<http://pandaflowers.net/aboutus.html>)

For the current year, Panda Flowers plans to plant one hectare of newly developed land with carnations. Like the roses being currently produced, the carnations will be exported to the USA.

The one-hectare plot is expected to be able to produce up to 50,000 stems per year, and Panda expects to be able to charge \$1.00 per stem. However, their customers will only pay for stems received in good condition, and based on past experience, Panda will sell (and thus be paid for) only 8 stems out of every 10 which are produced and shipped.

Under their current growing practices, Panda expects to have a variable cost (mostly labor and shipping, but also materials such as bulbs, pots, fertilizer, etc) of \$0.60 per stem produced.

Fixed costs will be approximately \$12,000 per year, and is mostly the cost of the land and equipment.

Required:

1. What is Panda's breakeven point for carnations under current growing practices (in number of stems produced per year)? Hint: it is more than their current production capacity.

$$\text{BEP } N^* = \frac{\text{Fixed Costs}}{\text{Unit sales price} - \text{Unit variable Costs}} = \frac{\text{Fixed Costs}}{\text{Unit Contribution margin}}$$

$$\rightarrow \text{Profit} = \underbrace{0.8 \times N^* \times \$1}_{\text{Revenue}} - \underbrace{\$0.60 \times N^*}_{\text{Variable Costs}} - \underbrace{\$12,000}_{\text{Fixed Costs}} = 0$$

$$N^* = \frac{\$12,000}{0.8 \times 1 - \$0.6} = 60,000 \text{ units (stems)}$$

One reason the initial plan (in part 1) is not feasible is that production is highly labor-intensive. Panda Flowers is considering an increase in automation to lower their variable labor costs, although this will also increase fixed costs (for example, by increasing rent and/or depreciation). Specifically, assume that for any reduction in variable cost per stem, there will be a 10,000 times greater increase in fixed costs per year. In addition, the reduction of variable cost is limited to a maximum of \$0.10 per stem - i.e., from \$0.60 per stem down to \$0.50 per stem. Further reduction below that is not possible regardless of the increase in fixed costs.

For example, to decrease variable costs by \$0.10, the largest possible reduction, fixed costs will increase by \$1,000 (10,000x\$0.10). To decrease variable costs by a tenth of a cent (\$0.001), fixed costs will increase by \$10 (10,000x\$0.001).

2. Assuming full production (50,000 stems) how much profit will Panda Flowers earn if it adopts the maximum amount of automation – this is, if it reduces variable costs by the full \$0.10 per stem?

$$\text{Profit} = \underbrace{0.8 \times 50,000 \times \$1}_{\text{Revenue}} - \underbrace{(\$0.6 - \$0.10) \times 50,000}_{\text{VC}} - \underbrace{\$12,000 + \$1,000}_{\text{FC}}$$

$$\text{Profit} = \$2,000$$

3. What is the minimum amount of automation (expressed in the reduction of variable cost per stem) that Panda Flowers must adopt in order to break even? Assume full production of 50,000 stems.

$$\text{Profit} = \underbrace{0.8 \times 50,000 \times \$1}_{\text{Revenue}} - \underbrace{(\$0.6 - x) \times 50,000}_{\substack{\text{VC} \\ \text{automation}}} - \underbrace{\$12,000 + 10,000x}_{\text{FC}} = 0$$

$$x = \$0.05$$

4. A large uncertainty about Panda Flowers' plan is the future demand for carnations. Assume that they cannot grow other flowers on the same plot of land, at least in the short term. Do you think that the automation plan makes the risk of future demand fluctuations more or less of a concern? Explain briefly, using the concept of operating leverage. No numbers are necessary!

More of a concern

Operating leverage - example

Question: compute the degree of operating leverage of the two firms. For each of the firms the contribution margin and operating income are given below. How would you interpret the results?

	Firm 1		Firm 2
Contribution margin	\$80,000	higher FC	\$20,000
Operating income	<u>\$10,000</u>		<u>\$10,000</u>
	$OL = \frac{\$80,000}{\$10,000} = 8$		$\frac{\$20,000}{\$10,000} = 2$

$$OL = \frac{\text{Total Contribution margin}}{\text{Operating income}} = \frac{\text{Contribution margin}}{\text{Contribution margin} - \text{Fixed Costs}}$$

An increase (decrease) of 1% in Sales, for Firm 1 results in 8% increase (decrease) in operating income. For Firm 2, an increase (decrease) of 1% in Sales results in 2% increase (decrease) in operating income.

Question 2 - Zeus Optical

Zeus Optical is a specialist manufacturer of optical instruments. Zeus has recently expanded its core product market of binoculars into making eyepieces for microscopes/telescopes, and screw-on lenses for digital SLR cameras. The firm believes that it makes little money selling binoculars, and that these new markets have great profit potential.

Somewhat to Zeus's surprise, it finds it tough to make money with eyepieces. As of now, the firm is selling the product at a negative profit margin. Yet, Zeus faces intense price pressure in this segment and thinks that it might have to lower prices by 5% or more to stay competitive. The market for binoculars has been stable for several years, and Zeus expects the trends to continue for the near future. Zeus is most excited about the market for screw-on lenses for digital SLR cameras. Although current volumes are small (relatively), Zeus believes that there is substantial market potential for this product. Leveraging its excellent reputation for optics and lenses, Zeus believes that it could reach and sustain three times the current volume of this product. This strategy also makes sense financially as this product is the most profitable of the three lines, per the firm's accounting records.

The following table provides key information about the product lines.

	<i>Eyepieces</i>	<i>Binoculars</i>	<i>Camera Lens</i>
Sales volume (units)	16,000	20,000	3,000
Price	\$53.00	\$78.00	\$160.00
Unit variable cost	30.00	45.00	118.00
Unit contribution margin	\$23.00	\$33.00	\$42.00
Unit profit margin	(\$2.50)	\$0.70	\$6.30
Labor hours/unit	1.5	1.9	2.1

Currently, the firm incurs \$1,161,100 in overhead costs annually. It allocates this overhead among product lines using the number of labor hours used by each product line. Zeus's management realizes that moving to camera lenses is a major shift in their product and market focus. Moreover, they know that factory personnel have complained about the increased coordination required for producing lenses. Thus, management wants you to conduct a detailed study of product costs.

You collect the following data.

	<i>Eyepieces</i>	<i>Binoculars</i>	<i>Camera Lenses</i>
Sales volume (units)	16,000	20,000	3,000
batch size	3,200	1,000	200
# of batches	5	20	15
# of receiving transactions	20	35	50
# of products	1	1	1
# of components	2	6	20

Analyzing the \$1,161,100 of overhead costs, you discover the following:

<i>Item</i>	<i>Amount</i>	
Per unit labor hour related	\$341,500	} Volume related
Per unit machine hour related	273,200	
Production order	88,000	} Execution of production order related
First part inspection	100,000	
Inventory management	110,000	} Inventory management related
Receiving and shipping	46,250	
Parts administration	90,000	→ Parts admin. related
General administration	112,150	→ Not allocated
Total	\$1,161,100	

You are wondering how best to allocate these costs into cost pools. You settle on forming five pools.

- Volume-related costs, allocated to products using labor hours.
- Cost related to executing a production order (this would include first part inspections), allocated using the number of batches.
- Costs related to inventory management, receiving, and shipping. These costs would be allocated using the number of receiving transactions per product line (= number of components x number of receiving transactions).
- Costs related to parts administration, which will be equally shared by all product lines.
- Facility-level general administration costs. You decide not to allocate these costs to individual product lines, reasoning that these costs are not controllable at the product level.

Required:

1. Compute the amounts in the five cost pools and the rate per driver unit for each cost pool.

Item	Amount			
Per unit labor hour related	\$341,500			
Per unit machine hour related	273,200			
Production order	88,000			
First part inspection	100,000			
Inventory management	110,000			
Receiving and shipping	46,250			
Parts administration	90,000			
General administration	112,150			
Total	\$1,161,100			

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- Facility-level general administration costs. You decide not to allocate these costs to individual product lines, reasoning that these costs are not controllable at the product level.

Cost pool	Amount	Cost driver	Qty	Allocation
Volume related	$\$341,500 + \$273,200$ $= \$614,700$	Labor hours	$1.5 \times 16,000 + 1.3 \times 20,000$ $+ 2.1 \times 3,000$ $= 68,300$	$= \frac{614,700}{68,300}$ $= \$9.00$
Order related	$\$88,000 + \$100,000$ $= \$188,000$	Number of batches	$5 + 20 + 15$ $= 40$	$\frac{\$188,000}{40}$ $= \$4,700$
Inventory related	$\$110,000 + 46,250$ $= \$156,250$	Number of transactions/line	$2 \times 2 + 35 \times 6 + 50 \times 20$ $= 1,250$	$\frac{\$156,250}{1,250}$ $= \$125$
Product related (parts administration costs)	$\$90,000$	Number of products	3	$\frac{\$90,000}{3}$ $= \$30,000$
Facility level cost	$\$112,150$	N/A	N/A	N/A

2. Using the driver rates you computed in part (a), determine the product cost for each product line. That is, compute the profit margin for each product line using activity-based costing.

	<i>Eyepieces</i> 16,000	<i>Binoculars</i> 20,000	<i>Camera Lens</i> 3,000
Sales volume (units)	16,000	20,000	3,000
Labor hours/unit	1.5	1.9	2.1
batch size	3,200	1,000	200
# of batches	5	20	15
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	<i>Eyepieces</i> 16,000	<i>Binoculars</i> 20,000	<i>Camera Lens</i> 3,000
Sales volume (units)	16,000	20,000	3,000
Price	\$53.00	\$78.00	\$160.00
Unit variable cost	30.00	45.00	118.00
Unit contribution margin	\$23.00	\$33.00	\$42.00
Volume related	$\$9 \times 1.5 = \13.5	$\$9 \times 1.9 = \17.10	$\$9 \times 2.1 = \18.9
order related	$\$4,700 / 3,200 = \1.47	$\$4,700 / 1,000 = \4.7	$\$4,700 / 200 = \23.5
Inventory related	$\$125 \times 2 \times 20 / 16,000 = \0.31	$\$125 \times 6 \times 35 / 20,000 = \1.31	$\$125 \times 20 \times 50 / 3,000 = \41.67
Product related	$30,000 / 16,000 = \$1.88$	$30,000 / 20,000 = \$1.5$	$30,000 / 3,000 = \$10$
Profit margin	\$5.84	\$8.39	\$(52.07)

To be computed on total basis ▽ Important

	<i>Eyepieces</i>	<i>Binoculars</i>	<i>Camera lenses</i>
Price	$\$53 \times 16,000 = \$848,000$	$\$78 \times 20,000 = \$1,560,000$	$\$160 \times 3,000 = \$480,000$
Variable costs	$\$30 \times 16,000 = \$480,000$	$\$45 \times 20,000 = \$900,000$	$\$118 \times 3,000 = \$354,000$
Contribution margin	\$368,000	\$660,000	\$126,000
Volume related	$\$13.5 \times 16,000 = \$216,000$	$\$17.10 \times 20,000 = \$342,000$	$\$18.90 \times 3,000 = \$56,700$
Order related	$\$1.47 \times 16,000 = \$23,520$	$\$4.7 \times 20,000 = \$94,000$	$\$23.5 \times 3,000 = \$70,500$
Inventory related	$\$0.31 \times 16,000 = \$5,000$	$\$1.31 \times 20,000 = \$26,200$	$\$41.67 \times 3,000 = \$125,000$
Product related	$\$1.88 \times 16,000 = \$30,080$	$\$1.5 \times 20,000 = \$30,000$	$\$10 \times 3,000 = \$30,000$
Profit margin	\$93,500	\$167,700	\$(156,200)

3. Based on the ABC profit margins, what actions would you recommend for Zeus's management.

Camera lenses : loss making , increasing volume will hurt the Company's profitability further . The price may be raised

Eyepieces : Profitable enough that the Company can price-Compete .

Binoculars : Profitable , but the company may reduce the number of orders and number of receiving transactions to reduce costs (same remark for camera lenses) .

Question 3. Bronson Company

Bronson Company manufactures a variety of ballpoint pens. The company has just received an offer from an outside supplier to provide the ink cartridge for the company's Zippo pen line, at a price of \$0.48 per dozen cartridges. The company is interested in this offer, since its own production of cartridges is at capacity.

Bronson Company estimates that if the supplier's offer were accepted, the direct labor and variable overhead costs of the Zippo pen line would be reduced by 10% and the direct materials cost would be reduced by 20%.

Under present operations, Bronson Company manufactures all of its own pens from start to finish. The Zippo pens are sold through wholesalers at \$4 per box. Each box contains one dozen pens. Fixed overhead costs charged to the Zippo pen line total \$50,000 each year. (The same equipment and facilities are used to produce several pen lines.) The expected cost of producing one dozen Zippo pens (one box) is given below:

Direct materials.....	\$1.50
Direct labor	1.00
Manufacturing overhead.....	<u>0.80*</u>
Total cost.....	<u>\$3.30</u>

*Includes both variable and fixed manufacturing overhead, based on production of 100,000 boxes of pens next year

Required:

- Should Bronson Company accept the outside supplier's offer? Show computations.

We have 100,000 boxes, with Total fixed costs of \$50,000
 Per unit (per box), Fixed OH = $\$50,000 / 100,000 = \0.5
 $\rightarrow$ Per box, Variable OH = $\$0.8 - \$0.5 = \$0.3$

Reduction in Cost when accepting the offer.

Direct materials	$\$1.5 \times 20\%$	\$0.3
Direct labor	$\$1 \times 10\%$	\$0.10
MOH (Variable)	$\$0.3 \times 10\%$	\$0.03
Total saving		<u>\$0.43</u>

Supplier's offer = \$0.48

We reject supplier's offer.

2. What is the maximum price that Bronson Company would be willing to pay the outside supplier per dozen cartridges?

Maximum price to pay = $\underline{\$0.43}$

3. Due to the bankruptcy of a competitor, Bronson Company expects to sell 200,000 boxes of Zippo pens next year. As stated above, the company presently has enough capacity to produce the cartridges for only 100,000 boxes of Zippo pens annually. By incurring \$30,000 in added fixed cost each year, the company could expand its production of cartridges to satisfy the anticipated demand for Zippo pens. The variable cost per unit to produce the additional cartridges would be the same as at present. Under these circumstances, should all 200,000 boxes of cartridges be purchased from the outside supplier, or should some or all of the 200,000 boxes of cartridges be made by Bronson? Show computations to support your answer.

Option 1: Producing 100% in-house

Variable costs: 200,000 boxes $\times$ \$0.43 per box

\$ 86,000

Fixed costs:

\$ 30,000

\$ 116,000

Option 2: Purchasing 100% from supplier

Variable costs: 200,000 boxes $\times$ \$0.48 per box

\$ 96,000

Option 3: Producing 50% in-house and purchasing 50% from supplier

Variable costs: 100,000 $\times$ \$0.43

= \$ 43,000

100,000 $\times$ \$0.48

48,000

\$ 91,000

Option 3 is the best.

Q. 4 – Olin Apparel Company

The Olin Apparel Company is a large producer of fabrics and garments under the "Arch" label. The company buys yarn on the market and converts it into fabric from which the garments are made. Two square meter of fabric is required to produce one garment. The costs and revenues associated with one unit of garment are given below:

		Per Garment
Selling price.....		\$30.00
Cost to manufacture:		
Raw materials:		
Buttons, thread, lining, etc.....	\$ 2.00	
Fabric	<u>16.00</u>	
Total raw materials	18.00	
Direct labor	5.80	
Manufacturing overhead.....	<u>8.70</u>	<u>32.50</u>
Manufacturing profit (loss).....		<u>\$ (2.50)</u>

Originally, all of the fabric was used to produce garments, but in recent years a market has developed for the fabric itself. The fabric is purchased by other companies for use in production of their own branded garments. Since the development of the market for the fabric, a continuing debate has persisted in the Olin Apparel Company as to whether the fabric should be sold simply as fabric or converted into garments. Current cost and revenue data on the fabric are given below:

		Per Sq. Meter of Fabric
Selling price.....		\$10.00
Cost to manufacture:		
Raw materials (e.g., yarn).....	\$ 3.50	
Direct labor	1.80	
Manufacturing overhead.....	<u>2.70</u>	<u>8.00</u>
Manufacturing profit/sq. meter.....		<u>\$ 2.00</u>

The market for garments is temporarily depressed, due to challenging economic conditions. This has made it necessary for the company to discount the selling price of the garments to \$30 from the normal \$40 price. However, the market for fabric has remained strong, the debate has again surfaced over whether the fabric should be sold outright rather than converted into garments. The VP of fabric unit thinks that the production of garments should be suspended; she is upset about having to sell garments at a \$2.50 loss, when the fabric could be sold for a healthy profit. However, the VP of production is equally upset at the suggestion that he close down a large portion of the factory. He argues that the company is in the garment business, not the fabric business, and that the company should focus on its area of strength.

Due to the nature of the production process, virtually all of the manufacturing overhead costs are fixed. Overhead is assigned to products (fabrics and garments) on a basis of 150 percent of direct labor cost.

Required:

1. Would you recommend that the fabric be sold outright, or converted into garments? Show computations in good form to support your answer, and explain your reasoning. [Hint: compute incremental contribution margin of converting fabric into a garment.]

		Per Garment			Per Sq. Meter of Fabric
Selling price		\$30.00	Selling price		\$10.00
Cost to manufacture:			Cost to manufacture:		
Raw materials:			Raw materials (e.g., yarn)	\$ 3.50	
Buttons, thread, lining, etc.	\$ 2.00		Direct labor	1.80	
Fabric	<u>16.00</u>		Manufacturing overhead	<u>2.70</u>	<u>8.00</u>
Total raw materials	18.00		Manufacturing profit/sq. meter		<u>\$ 2.00</u>
Direct labor	5.80				
Manufacturing overhead	<u>8.70</u>	<u>32.50</u>			
Manufacturing profit (loss)		<u>\$ (2.50)</u>			

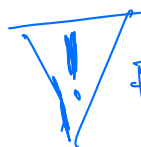
Additional revenue from further processing
the fabric into garment $\$ (30 - 2 \times \$10)$ \$10
Opportunity Cost

Additional costs from the processing
Buttons, thread, lining $\$2$
Direct labor $\$5.8$

\$7.8

Additional contribution margin

\$2.2



Fixed OH costs irrelevant to the decision making.

2. What is the lowest price that the company should accept for a garment? Show computations to support your answer, and explain your reasoning.

\$2.2 additional contribution margin made by selling garment for \$30/unit.

It means the lowest price the company should accept is
 $\$30 - \$2.2 = \$27.8$

Q. 5 – Variance Analysis - Acro-Cel Tool and Machine

During 2021, the Malaysia Division of Acro-Cel Tool and Machine had specialized in the production of a single heavy-duty motorcycle component. Original engineering estimates and the use of Acro-Cel's total overhead flexible budget had yielded the following standard cost per unit based on an expected production of 10,000 units:

STANDARD COST

Direct materials	10 lbs. @ \$1.50 per lb.	=	\$15.00
Direct labor	2 hours @ \$6.00 per hr.	=	12.00
Variable overhead	2 hours @ \$1.00 per direct labor hr.	=	2.00
Fixed overhead	2 hours @ \$4.00 per direct labor	=	<u>8.00</u>
Total standard cost			\$37.00

Tom, the Malaysia Division shop foreman, tends to communicate in brief, blunt reports. His analysis of 2021 is reproduced below:

As usual, the yo-yos in Purchasing and Personnel goofed and are trying to pass the buck to us. Your hotshot planners predicted an average wage rate of \$6.00, and then your bleeding heart labor negotiators conceded to \$6.60 an hour, for a 10% overrun. On top of that, Purchasing paid \$1.80 per pound for materials they should have gotten for \$1.50, or a whopping 20% over expenditure. Given that kind of incompetence, I feel the Malaysia production line deserves a big thank-you for holding actual total costs per unit down to \$41.58, which works out to 12.4% over standard.

ACTUAL EXPENDITURES:

Direct materials	\$226,800	Total manufacturing cost	\$498,920
Direct labor	\$166,320	Actual output	12,000 units
Variable overhead	\$24,800	Actual manufacturing	
Fixed overhead	\$81,000	cost per unit	\$41.58

REQUIRED

For ease of computation, assume that both the beginning and ending direct materials inventory balances were zero. Indeed, Acro-Cel uses a just-in-time inventory system, which means that no direct materials inventories are carried by Acro-Cel. Instead, materials are delivered to the factory couple of hours before use. Because labor is relatively plentiful, the production line is labor-intensive.

1. Tom's report is difficult to reconcile with a complaint from the controller's office that the Malaysia Division's cash expenditures were almost 35% higher than originally budgeted and had placed a severe strain on Acro-Cel's relationship with the bank. Rework Tom's report into a detailed variance analysis as best as you can. *Hint: you will need to calculate*

- The sales volume variance
- The sales price variance
- Material price variance.
- Material usage variance.
- Labor efficiency variance.
- Labor price variance, and
- Var. overhead. efficiency variance.
- Var. overhead spending variance, and
- Fixed overhead spending variance
- Production volume variance

Budget

Direct materials	10 lbs. @ \$1.50 per lb.	=	\$15.00
Direct labor	2 hours @ \$6.00 per hr.	=	12.00
Variable overhead	2 hours @ \$1.00 per direct labor hr.	=	2.00
Fixed overhead	2 hours @ \$4.00 per direct labor	=	8.00
Total standard cost			\$37.00

Actual

Direct materials	\$226,800	Total manufacturing cost	\$498,920
Direct labor	\$166,320	Actual output	12,000 units
Variable overhead	\$24,800	Actual manufacturing	
Fixed overhead	\$81,000	cost per unit	\$41.58

	^① Actual Results	^④ Price/Spending Variance	^① - ^③ - ^④ Efficiency Variance	^③ Flexible budget Actual $\times$ Budgeted P	^⑥ - ^③ - ^② Sales/Production Volume Variance	^② Static Budget
→ Units sold	12,000			12,000	2,000 F	10,000
→ Direct Materials	\$226,800	$126,000 \times (\$1.8 - \$1.5) = \$37,800 \text{ U}$	$(126,000 - 120,000) \times \$1.5 = \$9,000 \text{ U}$	$\$15 \times 12,000 = \$180,000$	$(12,000 - 10,000) \times \$15 = \$30,000 \text{ U}$	$\$15 \times 10,000 = \$150,000$
→ Direct Labor	\$166,320	$25,200 (\$6.6 - \$6) = \$15,120 \text{ U}$	$(25,200 - 24,000) \times \$6.00 = \$7,200 \text{ U}$	$\$12 \times 12,000 = \$144,000$	$(2 \times 12,000 - 2 \times 10,000) \times \$6 = \$24,000 \text{ U}$	$\$12 \times 10,000 = \$120,000$
→ Variable OH	\$24,800	$25,200 (\$1 - \$0.984) = \$400 \text{ F}$	$(25,200 - 24,000) \times \$1 = \$1,200 \text{ U}$	$\$2 \times 12,000 = \$24,000$	$(12,000 - 10,000) \times \$2 = \$4,000 \text{ U}$	$\$2 \times 10,000 = \$20,000$
→ Fixed OH	\$81,000	\$1,000 U		\$80,000	—	\$80,000
→ Total	\$498,920	\$53,520 U	\$17,400 U	\$428,000		\$370,000
		Total flexible budget variance \$70,920 U			Total sales volume variance \$58,000 U	

DM Consumed
\$226,800 / \$1.8 → The firm actually consumed 126,000 lbs

DL Consumed
\$166,320 / \$6.6 → The firm actually used 25,200 labor hours

OH (variable)
\$24,800 / 25,200 labor hours → 0.984 \$ per labor hour actually paid

2. Is Tom correct in his assessment of 2021 operating results? Be specific, supporting your analysis with as many numbers as possible.

No, even though there are unfavorable price / spending variances, the \$17,400 unfavorable efficiency variances are under Tom's control.